StyleRoute: Diffusion Style Transfer via Regional Routing and Conflict-aware Projection

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Abstract

Diffusion-based style transfer faces two fundamental questions: which regions should prioritize structure preservation and which are more amenable to texture transfer (the *where* question), and how to achieve such region-aware stylization without corrupting content (the *how* question). Existing methods typically rely on a single mechanism to control local structure, local texture, and global style together, making them fragile in region-rich scenes: uniform stylization strength may distort salient objects, while conservative control leaves texture-dominant regions under-stylized. We propose StyleRoute, a region-aware diffusion stylization framework that explicitly separates *where* style should be transferred from *how* the transfer should be optimized. For the *where* question, StyleRoute infers a region-level routing variable from structure confidence, texture confidence, and route disagreement, deciding per region whether to protect structure or to inject texture. For the *how* question, StyleRoute coordinates the regional routing with diffusion optimization to guide global style evolution, suppress style-content conflicts, and stabilize coarse layout. *Together, StyleRoute transforms the implicit style-content trade-off into an explicit, interpretable, and region-aware transfer process.* Experiments on diverse content-style pairs show consistent improvements over prior diffusion-based and region-matching stylization methods. Code will be released.

1 Introduction

Artistic style transfer aims to render a content image in the visual language of a reference style while preserving its semantic identity [1, 2, 3, 4]. In diffusion-based stylization, this goal becomes a *where-to-transfer* control problem rather than a simple content–style balancing task [5, 6, 7, 8]. Faces, animals, and building edges are sensitive to small geometric distortions, whereas foliage, fabric, and sky often require stronger appearance changes to appear stylized. The central question is therefore not only *how* to match content and style, but also *where structure should be protected and where texture should be transferred*, while keeping these local decisions compatible with a coherent global style.

Existing methods mostly improve the matching or injection mechanism, but they often leave these decisions coupled. Classical approaches align feature statistics [1, 2, 3]; attention-based models improve local correspondence [9, 4]; and diffusion methods manipulate latent features, self-attention, or semantic matches [5, 6, 7, 8]. Despite their differences, a single transfer strength is still expected to handle all regions uniformly, which leads to a clear failure: a strength that works for textured backgrounds often distorts faces or object boundaries, while a conservative setting that protects structure leaves texture-rich areas looking under-stylized, as shown in Figure 1.

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Figure 1: Motivation. A single implicit stylization strength is brittle: conservative transfer weakens painterly cues, while stronger transfer can damage fine structure.

To address this challenge, we propose StyleRoute, a diffusion stylization framework that explicitly separates *where* to transfer style from *how* to optimize the transfer. For *where*, StyleRoute reformulates stylization as a region-level routing problem. It decomposes stylization into a structure carrier and a texture carrier, and uses a region-wise stability gate to adaptively balance these two carriers according to local structural sensitivity and texture tolerance. When the two carriers conflict, the gate conservatively prioritizes structure preservation, since structural distortions are typically harder to recover than insufficient stylization.

For *how*, StyleRoute couples the above regional routing mechanism with global style evolution in diffusion. Through directional style alignment, sparse texture refinement, conflict-aware gradient projection, and a low-frequency layout anchor, StyleRoute encourages coherent image-level stylization while preventing style optimization from overwriting locally protected content structure. *As a result, the implicit style-content trade-off is transformed into an explicit, interpretable, and region-aware transfer process.*

This design is both effective and easy to analyze. The routing weights and projected gradients are explicit, so we can examine exactly what each component contributes. Experiments on the evaluated benchmark show that StyleRoute achieves a favorable balance between content preservation and stylization compared with the evaluated baselines. Our main contributions are three-fold:

- We formulate diffusion-based stylization as a regional routing problem, replacing a single uniform transfer strength with explicit structure and texture carriers.
- We introduce a stability-guided router that answers *where* to transfer, and a conflict-aware projection that answers *how* to transfer without hurting content structure.
- Experiments show that StyleRoute achieves a more favorable style–content trade-off than prior methods, improving across multiple evaluation metrics.

2 Related Work

2.1 Style Transfer and Diffusion-Based Stylization

Neural style transfer began with optimization-based matching of pretrained content features and style statistics [1, 10], but per-image optimization is slow and global statistics treat all regions uniformly. Feed-forward methods improved efficiency through perceptual losses and feature normalization [2, 3], while later methods used self-similarity, internal–external learning, contrastive objectives, and aesthetic pattern modeling for stronger local transfer [11, 12, 13]. Attention-based models such as AdaAttN [9] and StyTr^2 [4] further improved local correspondence with attention-weighted style statistics, yet they do not separate structure-sensitive regions from texture-tolerant ones. Diffusion methods expose disentangled style–content representations, inversion paths, h-space features, latent features, and attention maps for layout preservation, reference injection, semantic alignment, or training-free attention control [14, 15, 16, 5, 17, 6, 7, 18, 19]. Recent controllable stylization also studies structure–appearance separation [20]. Most methods still apply one transfer rule across the image; we instead route stylization at the region level.

The closest baselines are diffusion-based and semantic-aware stylization methods. Plug-and-play diffusion features [5] preserve spatial semantics through feature injection, but uses fixed injection

strength. StyleID [6] exposes the content–style trade-off through query preservation and attention temperature scaling, yet its temperature remains global. InstantStyle [7] uses lightweight reference conditioning from image-prompt adapters [21], but does not decide which regions receive structure or texture transfer. SCSA [22] and Semantix [23] improve region-level alignment through semantic sparse attention or energy-guided sampling. StyleGallery [8] is closest to our setting because it combines region segmentation, clustered matching, and regional style losses in a training-free semantic pipeline. These methods show the value of matching and semantic control, but to our knowledge, among these semantic stylization methods, none provides an explicit variable that routes each region between structure-preserving and texture-seeking transfer.

2.2 Routing, Balancing, and Conflict-Aware Optimization

Uncertainty-based weighting balances competing objectives through data-dependent allocation rather than fixed scalar weights [24]. In mixture-of-experts models, DSelect-k [25] studies smooth routing, and Switch Transformers [26] show that sparse expert selection scales efficiently. In multi-task learning, CAGrad [27] modifies updates under incompatible gradient directions, and PCGrad [28] projects conflicting gradients to reduce negative transfer. These lines suggest that a single weighted sum can hide useful structure when objectives interact. We apply these ideas selectively. The stability gate deterministically routes stylization mass between two carriers using confidence signals, rather than learning a separate expert. Conflict-aware projection likewise targets only the global style-content conflict, leaving the local routing branch untouched. This keeps each component simple and scoped to its specific role.

3 Method

3.1 Overview

Given a content image x_c and a style image x_s, the goal is to synthesize an output that preserves the structure of x_c while adopting the appearance of x_s. Our key idea is to treat style transfer as a problem of *where to transfer*: different regions of the content image have different capacities for accepting stylistic changes, and forcing uniform stylization often distorts salient objects while leaving texture-rich areas under-stylized. To address this, we introduce a region planning stage before diffusion. It segments the content image into N regions $\mathcal{R} = \{1, \dots, N\}$, each associated with a token-space mask m_r and area weight a_r, and retrieves from the style image a structure donor d_r^{str} and, when available, a texture donor d_r^{tex} for each region. This planning stage answers an initial question: *which regions should prioritize structure preservation, and which are more amenable to texture transfer*?

During diffusion, we optimize a latent z initialized from the content latent z_c. For the *where* decision, local structure routing pushes each region toward its structure donor, preventing over-stylization in salient areas. For the *how* decision, directional global style alignment guides overall appearance along the content-to-style residual, sparse texture refinement injects patterns only where needed, and conflict-aware projection removes the component of the style gradient that opposes content preservation. A low-frequency appearance anchor stabilizes coarse tone and layout throughout. Figure 2 summarizes the framework: regional routing decides *where* to transfer, while conflict-aware projection controls *how* to respect content constraints.

3.2 Where to Transfer: Region-Level Dual-Carrier Routing

3.2.1 Static Region Planning

Before denoising, StyleRoute builds a fixed region plan for the content image and the style candidate pool. The region masks are generated by automatic segmentation; DINOv2 supplies semantic region descriptors for representation and matching, but does not generate masks. The content and style images are decomposed into automatic regions, and each region is represented by a token-space mask, area, center, semantic prototype, and texture prototype, following the use of semantic region evidence in recent controllable stylization pipelines [8, 29]. The planner scores every content–style region pair twice: a structure score favors semantic and spatial compatibility, while a texture score favors reliable local texture transfer. The best role-compatible semantic match becomes the structure

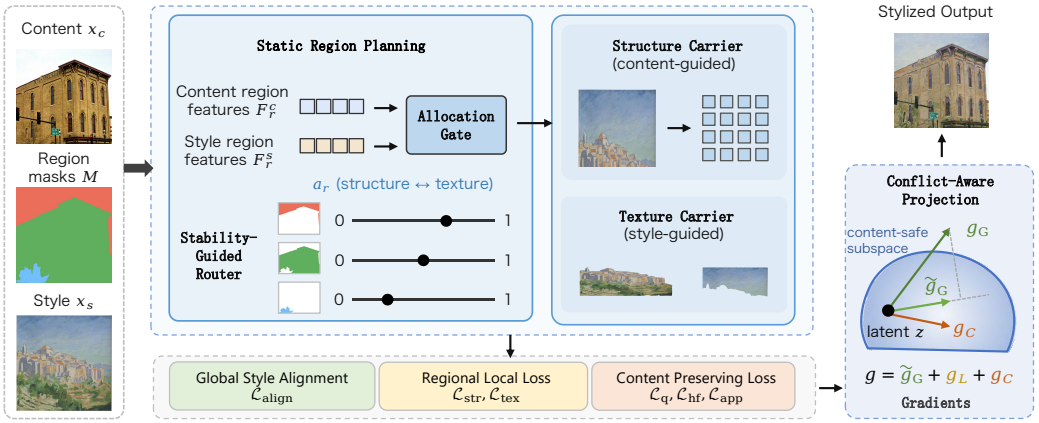


Figure 2: Method overview. The region-first stylization block answers *where* to transfer by estimating region-wise structure protection and texture eligibility. The global-control and diffusion-optimization blocks answer *how* to apply these regional decisions, using style alignment, conflict-aware projection, and appearance anchoring to preserve content structure.

donor d_r^{str}, and a texture donor d_r^{tex} is kept only when its texture score passes a routing threshold. The resulting masks, donors, confidence scores, and route disagreement values are fixed during diffusion and provide the inputs to the stability gate. Appendix A.1 gives the scoring definitions and implementation details.

3.2.2 Stability-Guided Dual-Carrier Routing

The main local decision is how much each region should rely on structure-preserving donor transfer rather than texture refinement. A uniform routing rule would treat a face and a patch of sky in the same way, although their tolerance to stylization is very different. Structural errors on salient objects are difficult to undo later, while an under-stylized sky can still absorb additional texture. We therefore define a region-specific stability gate g_r that favors structure when the evidence is ambiguous, using the same broad idea that data-dependent allocation can be preferable to a fixed scalar weight [24, 25, 26]:

$$g_r = \text{clip}_{[g_{\min}, g_{\max}]}\left(\eta_0 + \eta_s(\kappa_r^{(s)} - \kappa_r^{(t)}) + \eta_\delta \delta_r + b_r\right), \tag{1}$$

where η_0 is a base gate bias, $\kappa_r^{(s)}$ and $\kappa_r^{(t)}$ are structure and texture confidence, δ_r is route disagreement, b_r is a role prior for foreground-like regions, and $0 < g_{\min} < g_{\max} < 1$. The nonnegative coefficients η_s and η_δ encode two monotonic design constraints: stronger relative structure evidence and stronger route disagreement should not reduce structure protection. This conservative bias is useful because texture can still be added later, whereas geometric errors are hard to undo. The clipping interval $[g_{\min}, g_{\max}]$ prevents degenerate routing: every region keeps some structure protection, and no region is forced into pure structure transfer. The default gate uses $(\eta_0, \eta_s, \eta_\delta, g_{\min}, g_{\max}) = (0.48, 0.18, 0.22, 0.20, 0.85)$, with $b_r = 0.10$ for foreground-like regions and $b_r = -0.05$ otherwise.

The gate weights a mask-aware mixture of global style and region-specific donor responses. Let $o_{g,l}$ be the global style output at layer l, $o_{r,l}$ the donor-induced output for region r, and w_{bg} a background style-preservation weight. The structure target is

$$\hat{o}_l = \frac{w_{\text{bg}}\, o_{g,l} + \sum_{r\in\mathcal{R}} m_r(a_r g_r)\, o_{r,l}}{w_{\text{bg}} + \sum_{r\in\mathcal{R}} m_r(a_r g_r) + \varepsilon}, \qquad \mathcal{L}_{\text{str}} = \sum_l \|o_l(z) - \hat{o}_l\|_1. \tag{2}$$

Regions with large g_r are tied more strongly to structure donors, and the complementary weight $1 - g_r$ is used in the texture branch. We set $w_{\text{bg}} = 0.35$ for all experiments.

3.3 How to Optimize: Conflict-Aware Diffusion Stylization

3.3.1 Directional Global Style Alignment

Global alignment controls the image-level style direction, while gated sparse texture refinement controls local texture injection. Direct image-level style matching can pull the output toward a mismatched global style state even when each region receives a plausible donor, because locally reasonable assignments do not guarantee a coherent global style. We therefore measure style evolution relative to the content image and guide the diffusion trajectory along a controlled direction instead of forcing it to copy the style state exactly.

For each style-alignment layer $l \in \mathcal{S}_{\text{align}}$, a descriptor $\phi_l(\cdot)$ is constructed by spatially standardizing the attention output, applying mild smoothing, and pooling to a fixed grid. Let z_s be the VAE latent of the selected global style image, and let D denote the number of scalar entries in the flattened descriptor. The current and style residuals are

$$c_l = \phi_l(o_l(z)) - \phi_l(o_l(z_c)), \qquad s_l = \phi_l(o_l(z_s)) - \phi_l(o_l(z_c)). \tag{3}$$

Rather than forcing $c_l = s_l$, the current residual is decomposed into a component parallel to the style direction and an orthogonal residual:

$$u_l = \frac{s_l}{\|s_l\|_2 + \varepsilon}, \qquad p_l = \langle c_l, u_l \rangle, \qquad c_l^{\perp} = c_l - p_l u_l. \tag{4}$$

The style-alignment loss is

$$\mathcal{L}_{\text{align}} = \frac{1}{|\mathcal{S}_{\text{align}}|} \sum_{l \in \mathcal{S}_{\text{align}}} \left[\frac{|p_l - \rho \|s_l\|_2|}{\|s_l\|_2 + \varepsilon} + \lambda_{\perp} \frac{\|c_l^{\perp}\|_1}{D(\|s_l\|_2 + \varepsilon)} \right]. \tag{5}$$

The parallel term moves the current image along the style direction to a target magnitude set by ρ, while the orthogonal term suppresses drift in unsupported directions. We use $\rho = 0.65$ and $\lambda_{\perp} = 0.35$, yielding a controlled global style trajectory instead of direct style copying.

3.3.2 Sparse Texture Refinement

Texture transfer is restricted in both space and time. Dense texture injection across all regions often damages semantic anchors. Adding brushstroke patterns to a face, for example, can harm identity even when the local texture match is reasonable. We therefore activate texture refinement only for regions with reliable texture evidence, and only for the top-K regions ranked by the texture budget score:

$$b_r^{\text{tex}} = (1 - g_r)\kappa_r^{(t)} \sqrt{a_r} \max(0, 1 - \eta_{\delta}^{\text{tex}} \delta_r), \qquad K = \min\big(K_{\max}, \lceil \sqrt{N_{\text{tex}}} \rceil\big), \tag{6}$$

where N_{tex} is the number of regions with texture donors. This score favors regions with strong texture evidence, sufficient support, and low route disagreement; we use $\eta_{\delta}^{\text{tex}} = 0.5$ and $K_{\max} = 3$ to keep the branch sparse even in region-rich images. For each active region, high-pass filtered query features from texture layers $\mathcal{S}_{\text{tex}}$ are compared inside the masked region through a grid descriptor ψ_l:

$$\mathcal{L}_{\text{tex}} = \frac{1}{|\mathcal{A}|} \sum_{r \in \mathcal{A}} (1 - g_r) \frac{1}{|\mathcal{S}_{\text{tex}}|} \sum_{l \in \mathcal{S}_{\text{tex}}} \big\| \psi_l(q_l(z), m_r) - \psi_l(q_l(d_r^{\text{tex}}), m_r) \big\|_1. \tag{7}$$

Let $\pi_t = (t+1)/T$ denote normalized diffusion progress. The texture schedule is

$$\gamma_t = \mathbf{1}\{\pi_t > \tau_{\text{tex}}\} \left[1 + \beta_{\text{tex}} \operatorname{clip}_{[0,1]} \left(\frac{\pi_t - \tau_{\text{tex}}}{1 - \tau_{\text{tex}}} \right) \right]. \tag{8}$$

We set $(\tau_{\text{tex}}, \beta_{\text{tex}}) = (0.55, 0.35)$, so γ_t keeps this branch inactive in early diffusion steps and raises its weight only after the global layout has largely stabilized.

3.3.3 Conflict-Aware Projection with Appearance Anchoring

Appearance Anchoring. The content branch preserves content features through query matching and high-frequency query matching. Let $q_l^c = q_l(z_c)$ be the content query feature at layer l, $\text{sg}(\cdot)$

stop gradients through the reference feature, and $\mathcal{H}_{\sigma_q}$ a token-grid high-pass filter. We use

$$\begin{aligned}\mathcal{L}_q &= \sum_l \left\| q_l(z) - \mathrm{sg}(q_l^c) \right\|_1, \\ \mathcal{L}_{\mathrm{hf}} &= \sum_l \left\| \mathcal{H}_{\sigma_q}(q_l(z)) - \mathrm{sg}\big(\mathcal{H}_{\sigma_q}(q_l^c)\big) \right\|_1 .\end{aligned} \tag{9}$$

We also add a low-frequency appearance anchor to stabilize coarse tone:

$$\mathcal{L}_{\mathrm{app}} = \left\| \mathcal{G}_\sigma\big(\mathcal{V}(z)\big) - \mathcal{G}_\sigma(x_c) \right\|_1 , \tag{10}$$

where $\mathcal{V}$ is the differentiable VAE decoder and $\mathcal{G}_\sigma$ is Gaussian blur. We use $\sigma = 5.0$; because the blur removes high-frequency details, this term constrains low-frequency appearance without suppressing fine stylization. In practice, it mainly prevents slow drift in overall tone and layout.

Conflict-Aware Projected Update. Based on the previously defined individual losses, the unprojected scalar combination is

$$\mathcal{L} = \alpha_t \mathcal{L}_{\mathrm{align}} + \mathcal{L}_{\mathrm{str}} + \lambda_{\mathrm{tex}} \gamma_t \mathcal{L}_{\mathrm{tex}} + \lambda_c \left(\mathcal{L}_q + \lambda_{\mathrm{hf}} \mathcal{L}_{\mathrm{hf}} \right) + \lambda_{\mathrm{app}} \mathcal{L}_{\mathrm{app}}, \tag{11}$$

where $\alpha_t = 0.10 + 0.45(1 - \pi_t)^2$ decays quadratically to emphasize global guidance early, and γ_t activates texture refinement after a fraction τ_{tex} of the diffusion trajectory. The loss weights are fixed to $\lambda_{\mathrm{tex}} = 0.12$, $\lambda_c = 0.25$, $\lambda_{\mathrm{hf}} = 0.10$, and $\lambda_{\mathrm{app}} = 0.08$. The actual update is not the plain gradient of this scalar combination, because the global style term can conflict with content preservation. These losses guide per-image latent optimization during diffusion inference only; all pretrained diffusion-model parameters remain frozen, without training or fine-tuning.

The main optimization conflict appears between the global style-alignment term and content preservation. The style-alignment loss can improve style coherence while pushing the latent away from content-consistent solutions. A fixed scalar weight does not resolve this conflict reliably because the conflict direction changes across the latent space and over diffusion steps, which is the setting addressed by gradient-conflict methods in multi-objective optimization [27, 28]. We therefore decompose the latent gradient into three components:

$$\begin{aligned} g_G &= \nabla_z(\alpha_t \mathcal{L}_{\mathrm{align}}), \\ g_L &= \nabla_z(\mathcal{L}_{\mathrm{str}} + \lambda_{\mathrm{tex}} \gamma_t \mathcal{L}_{\mathrm{tex}}), \\ g_C &= \nabla_z\big(\lambda_c(\mathcal{L}_q + \lambda_{\mathrm{hf}} \mathcal{L}_{\mathrm{hf}}) + \lambda_{\mathrm{app}} \mathcal{L}_{\mathrm{app}}\big). \end{aligned} \tag{12}$$

When the global and content gradients conflict ($\langle g_G, g_C \rangle < 0$), we project g_G to suppress the component that opposes g_C:

$$\begin{aligned} \tilde{g}_G &= g_G - \frac{\langle g_G, g_C \rangle}{\|g_C\|_2^2 + \varepsilon} g_C, \\ g &= \tilde{g}_G + g_L + g_C. \end{aligned} \tag{13}$$

Here, only the global branch is projected. The local branch is already moderated by the region gate, whereas the global style-alignment term is the main source of image-level drift. Under a conflict, the projection suppresses the component of g_G along $-g_C$ up to the numerical ε, so the global style update no longer directly cancels the content-preserving branch before the local gradient is added.

4 Experiments

4.1 Experimental Setup

Datasets and Settings. Following prior style transfer protocols [6, 4, 30], we randomly sample 30 content images from MS-COCO [31] and FFHQ [32] and 30 style images from WikiArt [33]. The sampled images are paired into a fixed held-out evaluation set, and all methods use the same content–style pairs. The content pool contains both region-rich scenes and structure-sensitive faces, so the evaluation tests object layout, hard boundaries, facial geometry, and texture-heavy backgrounds.



Figure 3: Qualitative comparison with baselines. StyleRoute preserves structure-sensitive geometry across diverse scenes while retaining visible style texture. Best viewed zoomed in.

Table 1: Comparison on the held-out random content–style evaluation set. Best and second-best results are bold and underlined.

Method	ArtFID↓	FID↓	LPIPS↓	CFSD↓	B-Gram ($\times 10^{-3}$)↓
StyTR^2 [4]	35.30	23.69	0.4299	0.0608	<u>2.268</u>
StyleID [6]	33.34	20.14	0.5776	0.1683	3.424
CoCoDiff [30]	33.99	20.27	0.5984	0.1882	3.436
InstantStyle [7]	43.70	26.50	0.5890	0.2726	3.669
AD [36]	30.21	<u>19.49</u>	0.4743	0.0759	2.492
CSGO [35]	33.37	20.85	0.5269	0.1643	3.044
StyleGallery [8]	<u>28.90</u>	19.51	0.4090	0.0602	2.314
StyleGallery+SAM [8]	28.97	19.56	<u>0.4087</u>	<u>0.0601</u>	2.314
StyleRoute (Ours)	**28.29**	**19.11**	**0.4070**	**0.0551**	**2.190**

Implementation Details. We compare StyleRoute with eight baselines: StyleGallery [8], StyleGallery with its released SAM option enabled [34], StyTR^2 [4], StyleID [6], CoCoDiff [30], InstantStyle [7], CSGO [35], and Attention Distillation (AD) [36]. StyleGallery is the direct semantic region-matching reference; StyleGallery+SAM isolates mask access by changing only the released mask option. The remaining baselines cover transformer feed-forward transfer, explicit content–style balancing, correspondence-aware diffusion, SDXL image-prompt/adapters, and attention manipulation. All baselines use public code and recommended settings, including default diffusion backbones. For StyleRoute, latent optimization uses Adam [37] under a DDIM scheduler [38] on a single NVIDIA A6000 GPU. We use one *fixed* scalar configuration for all content–style pairs; key parameter settings are summarized in Appendix A. The effect of changing the routing parameter is analyzed later in the gate-sensitivity study.

Metrics. We report ArtFID [39], FID [40], LPIPS [41], CFSD [6], and block-Hungarian Gram distance (B-Gram). The computation of B-Gram is detailed in Appendix A.3. These metrics cover style-domain quality, content and structural preservation, and style-statistic alignment; lower is better for all. Reporting all five is important because gains on one side of the trade-off can degrade the other, as discussed in Section 2.

4.2 Comparison with SOTAs

Qualitative Comparison. Figure 3 supports the quantitative trends. On animals, clocks, and other structure-sensitive regions, StyleRoute preserves geometry more reliably than StyleID and StyleGallery while avoiding the overly smooth appearance often produced by AD. On texture-dominant foliage, walls, and sky, it still introduces visible brushstroke and surface patterns. The gain is clearest on mixed scenes: regional routing estimates where texture can be absorbed and where structure should stay anchored, keeping object geometry stable while producing textured backgrounds.

Quantitative Comparison. Table 1 reports the quantitative comparison against existing style transfer methods. StyleRoute achieves the best performance across all five metrics, including ArtFID, FID, LPIPS, CFSD, and B-Gram. The baselines exhibit two recurring failure modes: conservative methods (e.g., StyleGallery, AD) preserve content structure but leave weak style alignment, while aggressive methods (e.g., StyTR^2, InstantStyle) push stylization at the cost of content distortion. By explicitly allocating style strength per region and resolving gradient conflicts globally, StyleRoute avoids both failure modes and attains a more favorable style–content operating point.

We further compare against two region-aware variants of StyleGallery [8]: the released no-SAM version and the SAM-augmented version. Adding SAM to StyleGallery yields nearly identical performance across all metrics, indicating that simply injecting segmentation masks without a principled routing mechanism does not effectively exploit region-level information. In contrast, StyleRoute incorporates regional routing and conflict-aware projection, achieving consistent improvements over both variants. This gap demonstrates that region awareness alone is insufficient; the key lies in *how* region information guides the allocation of structure preservation and style transfer strengths.

An additional paired evaluation against StyleGallery, including confidence intervals and computational costs, is reported in Appendix B.1–B.2.

User Study. We conducted a perceptual study on 30 content–style groups with 50 participants. StyleRoute received 62.3% preference votes, outperforming StyleGallery (17.7%), AD (11.0%), and StyleID (9.0%).

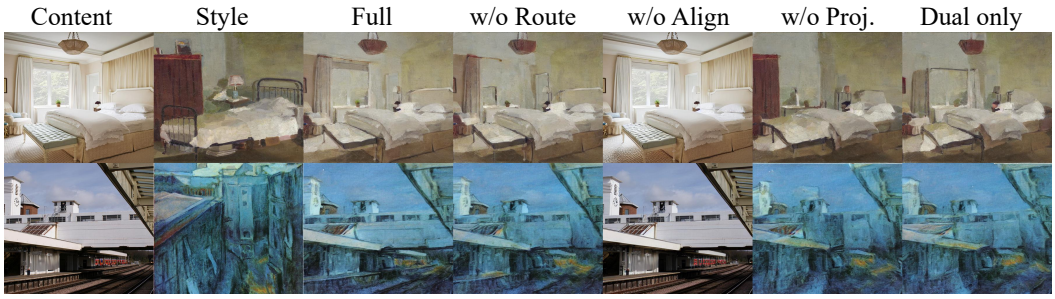


Figure 4: Qualitative ablation. Columns show content, style, full model, three removed-component variants, and dual-carrier-only transfer.

4.3 Model Analysis

Ablation Study. Figure 4 provides visual evidence for the component ablations. In the figure, *Content* and *Style* are the input content and reference style images, and *Full* is the complete StyleRoute model. *w/o Route* removes the stability-guided regional routing decision from the local branch; *w/o Align* removes the directional style-alignment loss $\mathcal{L}_{\text{align}}$; *w/o Proj.* keeps the losses but removes conflict-aware gradient projection; and *Dual only* keeps the dual-carrier local transfer branch without the global controls. The removed-component variants show that the ablations move along the style–content trade-off rather than only changing the apparent strength of stylization. This supports the interpretation that explicit regional decision points help separate structure preservation from texture transfer.

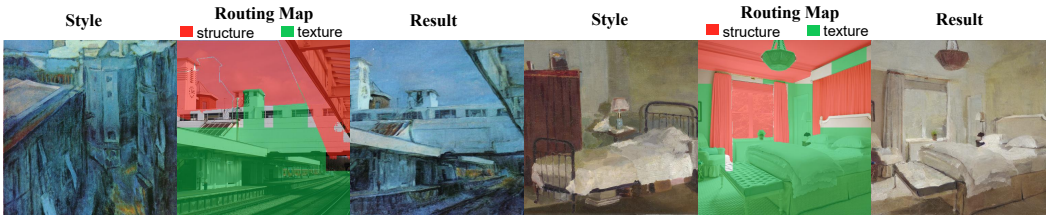


Figure 5: Routing maps from the stability gate: red denotes structure-protected regions and green denotes texture-eligible regions, revealing spatially selective transfer rather than a single global stylization strength.

Routing Maps. Figure 5 visualizes the gate-induced where-to-transfer allocation, showing that protected structure and texture-eligible regions are separated spatially rather than controlled by one global stylization strength.

Table 2: Ablation and gate diagnostics. Left: ablation study on routing and global controls; B-Gram is reported in 10^{-3} and lower is better. Right: gate-intercept diagnostic on a representative room case with a fixed cached region plan; increasing η_0 raises the structure gate and area-weighted structure mass $\sum_r a_r g_r$ while reducing the complementary texture budget $\sum_r b_r^{tex}$.

Ablation operating points

Config.	LPIPS↓	CFSD↓	B-Gram↓
StyleRoute	0.4070	**0.0551**	**2.19**
w/o routing	0.5809	0.0925	2.31
w/o style align	**0.3805**	0.0847	3.55
w/o projection	0.6829	0.1221	2.30
Dual-carrier	0.6240	0.1409	2.30

Gate-intercept diagnostic

η_0	mean g	FG	BG	Struct.	Tex.
0.36	0.389	0.465	0.313	0.354	1.312
0.42	0.449	0.525	0.373	0.411	1.185
0.48	0.509	0.585	0.433	0.467	1.058
0.54	0.569	0.645	0.493	0.524	0.931
0.60	0.629	0.705	0.553	0.581	0.804

Ablation Operating Points. The ablations reproduce the same two failure modes observed among the completed baselines. Removing style alignment collapses the optimization toward content reconstruction: content distances become small, but style-domain quality and style statistics degrade. Removing routing, projection, or the global controls moves in the opposite direction: style-domain metrics become more aggressive, but content and structural distances worsen. Mechanistically, style alignment keeps global style evolution active, routing decides which regions can absorb texture, and conflict-aware projection is designed to suppress the component of the global style gradient that opposes content preservation. The full model is therefore not designed to minimize one isolated metric; it is designed to mitigate both under-stylization and structure damage, which is the operating point supported by Table 2.

Budget-matched Uniform and Shuffled routing controls are reported in Appendix B.3 to isolate the effect of spatial allocation.

Gate Sensitivity. Table 2 isolates the intercept of the Stability Gate while keeping the cached region plan fixed. Increasing η_0 translates both foreground and background gates upward, which increases the area-weighted structure mass and reduces the complementary texture budget. The foreground–background gap remains nearly constant because it is controlled by the role prior rather than by η_0. This diagnostic shows that the gate coefficients act as interpretable routing controls with predictable structure–texture semantics, rather than as brittle hand-coded constants.

5 Conclusion

This paper presents StyleRoute for diffusion-based style transfer through regional routing and conflict-aware projection. Different regions need different amounts of structure protection and texture transfer, while global style updates must not cancel content preservation. StyleRoute implements this idea through stability-guided regional routing, directional style alignment, conflict-aware projection, and a low-frequency appearance anchor. Across the compared baselines under our evaluation protocol,

StyleRoute improves the measured style-domain and content-preservation metrics over the closest semantic matching baseline. This suggests that regional routing and projected global updates can improve the measured operating point, rather than merely shifting a fixed style–content weight. Despite such achievements, the current gate relies on hand-crafted confidence signals, and learned routing may adapt better to unfamiliar scene layouts. Besides, donor retrieval is still limited by the style candidate pool, so better retrieval could improve both structure and texture donors.

Performance depends on segmentation and donor quality, and the evaluated non-painting categories show mixed content–style trade-offs (Appendix B.4). The method incurs additional inference-time cost, and its compatibility with DiT/MMDiT backbones remains unvalidated.

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A Implementation Details

A.1 Region Planning

The static region plan uses automatic masks. For each region, its semantic descriptor is the normalized masked mean of DINOv2 tokens, and its texture descriptor concatenates the masked mean and standard deviation of U-Net features. The area a_r and center μ_r are computed from the image-space mask. For content region r and style region j of candidate i, the compatibility terms are

$$\begin{aligned} D_{rij} &= \langle p_r^{\mathrm{sem}}, p_{ij}^{\mathrm{sem}} \rangle, & T_{rij} &= \langle p_r^{\mathrm{tex}}, p_{ij}^{\mathrm{tex}} \rangle, \\ P_{rij} &= \exp\!\left(-\frac{\|\mu_r - \mu_{ij}\|_2^2}{2\sigma_p^2} \right), & A_{rij} &= \exp\!\left(-\frac{|\log(a_r/(a_{ij}+\varepsilon))|}{\sigma_a} \right). \end{aligned} \tag{14}$$

After normalization over candidate regions, structure and texture compatibility scores are

$$\begin{aligned} s_{rij}^{\mathrm{str}} &= \omega_D D_{rij} + \omega_P P_{rij} + \omega_A A_{rij} + \omega_G \bar{s}_i^{\mathrm{cand}}, \\ s_{rij}^{\mathrm{tex}} &= \nu_T T_{rij} + \nu_D D_{rij} + \nu_A A_{rij} + \nu_G \bar{s}_i^{\mathrm{cand}}, \end{aligned} \tag{15}$$

where $\bar{s}_i^{\mathrm{cand}}$ is the normalized global candidate score. The structure donor is the highest-scoring role-compatible match. The texture donor is the highest-scoring candidate satisfying $s_{rij}^{\mathrm{tex}} \geq 0.14$; otherwise the texture branch is inactive for that region. The selected scores define structure/texture confidence. Route disagreement is the clipped complement of the semantic similarity between the selected donors (or 1 if no texture donor exists). All assignments are fixed before diffusion.

A.2 Key Parameters

Component	Setting used in the reported experiments
Diffusion optimization	Frozen U-Net; content-latent initialization; empty-prompt embedding; 150 DDIM steps; Adam learning rate 0.05; one NVIDIA A6000 GPU.
Routing gate	$(\eta_0, \eta_s, \eta_\delta) = (0.48, 0.18, 0.22)$; $[g_{\min}, g_{\max}] = [0.20, 0.85]$; foreground/background role priors $0.10/-0.05$.
Donor and structure	Texture donor threshold 0.14; tight/context reference blend 0.35; background support $w_{\mathrm{bg}} = 0.35$.
Global alignment	Middle and last self-attention layers; 8×8 descriptor; $(\rho, \lambda_\perp) = (0.65, 0.35)$.
Sparse texture	Top $K \leq 3$ regions; 4×4 descriptor; $\eta_\delta^{\mathrm{tex}} = 0.5$; $(\tau_{\mathrm{tex}}, \beta_{\mathrm{tex}}) = (0.55, 0.35)$.
Appearance / content	Query high-pass Gaussian $\sigma_q = 1.0$; appearance Gaussian $\sigma_{\mathrm{app}} = 5.0$.
Loss / projection	$(\lambda_{\mathrm{tex}}, \lambda_{\mathrm{c}}, \lambda_{\mathrm{hf}}, \lambda_{\mathrm{app}}) = (0.12, 0.25, 0.10, 0.08)$; $\varepsilon = 10^{-6}$.

Table 3: Implementation settings fixed across all evaluated content–style pairs.

A.3 B-Gram Evaluation Metric

For each VGG evaluation layer l, split output and style features into spatial blocks. With $F_l^b(x) \in \mathbb{R}^{C_l \times n_b}$, let $G_l^b(x) = F_l^b(x) F_l^b(x)^\top / (C_l n_b)$. A one-to-one Hungarian assignment $\pi_l^\star$ minimizes the sum of Frobenius distances between output and style block Gram matrices. The metric averages the matched distances across blocks and layers:

$$\mathrm{B\text{-}Gram}(x, s) = \frac{1}{|\mathcal{L}_G|} \sum_{l \in \mathcal{L}_G} \frac{1}{|\mathcal{B}_l|} \sum_{b \in \mathcal{B}_l} \left\| G_l^b(x) - G_l^{\pi_l^\star(b)}(s) \right\|_F. \tag{16}$$

A.4 Perceptual Preference Study

For 30 fixed content–style groups, 50 participants compared anonymized, randomly ordered outputs from StyleRoute, StyleGallery, AD, and StyleID alongside the content and style references. They selected the result best balancing content preservation and style transfer; aggregate preferences are reported in Section 4.2.

B Additional Quantitative Evidence

B.1 Expanded Paired Evaluation

To assess the stability of the comparison with StyleGallery, we evaluate an additional 500 content–style pairs fixed before evaluation, using identical inputs for both methods and reporting results separately from Table 1. For the mean paired difference (StyleRoute minus StyleGallery), 95% CIs are the 2.5th and 97.5th percentiles of 10,000 bootstrap resamples of pair indices with replacement; they reflect variation across pairs, not repeated runs.

Metric ↓	Mean paired difference	95% CI
LPIPS	-0.0158	$[-0.0216, -0.0102]$
CFSD	-0.0091	$[-0.0109, -0.0074]$
B-Gram	-0.000108	$[-0.000124, -0.0000917]$

Table 4: Additional 500-pair comparison: negative differences favor StyleRoute.

All three intervals lie below zero; ArtFID improves slightly, while FID remains essentially unchanged.

B.2 Computational Cost

Under the same measurement protocol, StyleRoute increases per-pair runtime by approximately 8.7% and peak GPU memory by 71.8% relative to StyleGallery. StyleRoute uses 9.26 GB of peak GPU memory. This reflects additional inference-time computation and memory for explicit regional control.

B.3 Controlled Spatial Routing

We compare Full routing with Uniform and Shuffled controls. All variants use the same masks, donors, aggregate structure/texture budgets, initialization, losses, and schedules. Uniform removes spatially varying routing while preserving the total budgets; Shuffled reassigns the routing coefficients to different regions. This intervention isolates the spatial allocation of routing coefficients.

Regions with the highest structure-routing weights in the Full plan define the protected-region evaluation, and Full texture-active masks define the texture-region evaluation. Table 5 reports the percentage changes from the controlled evaluation.

Metric	Full vs. Uniform	Full vs. Shuffled
Protected-region SSIM ↑	$+1.54\%$	$+1.35\%$
Protected-region boundary discrepancy ↓	-2.39%	-2.16%
Protected-region pixel L1 ↓	-0.72%	-0.84%
Texture-region donor-Gram ↓	-0.37%	-0.24%
LPIPS-content ↓	-2.27%	-2.49%
CFSD ↓	-3.75%	-3.43%

Table 5: Reported routing-control changes (Full minus control). Positive SSIM changes and negative error changes favor Full.

The comparisons support the role of spatial allocation in protected-region structure preservation and texture-region consistency, beyond simply providing region masks or aggregate transfer budgets.

B.4 Style-Domain Breadth and Mask Perturbations

We evaluate five fixed mountain content images with four style references in each of five non-painting categories (20 pairs per category; 100 total). Both methods use the same 150-step setting, without output filtering or replacement. The limited content diversity constrains generalization beyond these evaluated scenes.

Category	LPIPS-content	CFSD	B-Gram
Anime/comic	-0.04679	-0.02112	$+3.04\times10^{-4}$
Graphic/poster	-0.02832	-0.03864	$+2.33\times10^{-4}$
Photographic grading	$+0.06029$	$+0.04449$	-3.88×10^{-4}
Pixel art	-0.02394	-0.10088	-2.41×10^{-4}
Sketch/engraving	$+0.04007$	$+0.01155$	-1.52×10^{-4}
All 100 pairs	$+0.000263$	-0.02092	-4.89×10^{-5}

Table 6: Non-painting evaluation: StyleRoute minus StyleGallery; negative values favor StyleRoute for all three metrics.

Results are mixed: photographic grading and sketch/engraving improve B-Gram but worsen LPIPS-content and CFSD, while anime/comic and graphic/poster show the opposite trade-off. Aggregate LPIPS is approximately tied. These findings do not establish uniformly better cross-domain performance.

For segmentation robustness, we test one-step erosion, one-step dilation, and deterministic two-pixel mask displacement while holding image pairs and optimization settings fixed. Across the reported LPIPS-content, CFSD, B-Gram, protected-region L1, and donor-Gram endpoints, the absolute relative changes are at most 0.32%. This supports stability under the tested mild geometric perturbations, but does not establish robustness to missing objects, severe segmentation errors, or incorrect donor assignments.

C Additional Qualitative Results

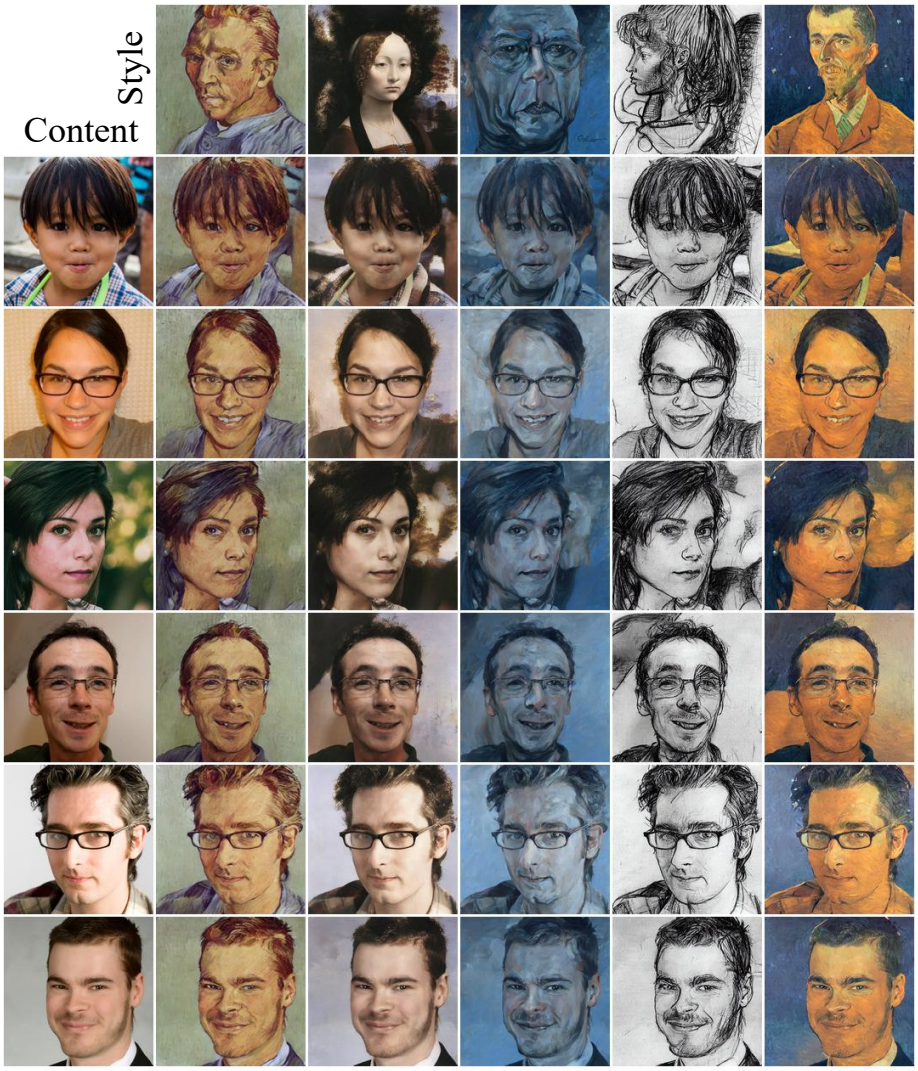


Figure 6: Additional style transfer results on portraits.



Figure 7: Additional style transfer results on buildings.

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